Universität Zürich[UZH]
ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

DaST
Data • (Systems+Theory)
VLG Computer Vision and Learning Group

Bachelor's Thesis

Submission of the thesis
April 17, 2026

Advancing (Sparse) Multi-View Depth Estimation

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Abstract

Zusammenfassung

Acknowledgements

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1

Introduction

Recovering the three-dimensional structure of a scene from visual observations is a foundational problem in computer vision, with applications spanning robotics, autonomous navigation, and augmented reality [4]. In particular, metric depth estimation enables downstream systems to reason about geometry and spatial relationships in real environments. While deep learning has substantially improved monocular depth estimation [3, 1], extending these methods to sparse multi-view settings remains a challenging and practically important problem.

When multiple synchronized cameras observe the same scene over time, the reconstruction problem extends naturally from static 3D geometry to modeling temporal scene evolution - commonly referred to as 4D reconstruction. Here the objective is not only to recover accurate per-frame geometry, but to ensure that the recovered structure is consistent across the sequence. Achieving both geometric accuracy and temporal stability under sparse camera configurations (typically two to four cameras, as encountered in robotic platforms) is the central problem of this thesis.

1.1 Background and Motivation

Classical approaches to multi-view reconstruction, such as Structure-from-Motion (SfM) [7] and Multi-View Stereo (MVS) [8], rely on sparse feature correspondences and struggle in scenes with textureless regions, dynamic objects, or limited viewpoint overlap. Recent learning-based methods have largely superseded these pipelines by reformulating reconstruction as dense prediction. DUSt3R [10] and MASt3R [6] in particular demonstrate strong robustness in sparse-view settings by directly regressing pointmaps from image pairs, bypassing explicit camera calibration. MASt3R-SfM [2] extends this framework to full scene reconstruction through global pointmap alignment, whereas VGGT [9] employs a joint optimization of camera parameters and dense geometry across multi-view collections to ensure global consistency.

Needs citation

These spatial methods are typically evaluated on a per-frame basis, and their behavior in the temporal domain is less well understood. Methods designed specifically for dynamic scenes - such as VGGT4D [5] and Mono4D [11] - incorporate temporal information to encourage smoother reconstructions, but may trade off spatial precision in doing so. Whether methods that excel under standard per-frame benchmarks also produce temporally stable 4D reconstructions remains an open and underexplored question.

1.2 Problem Statement and Contributions

This thesis investigates metric 4D reconstruction from sparse multi-view video. Given synchronized video streams from two to four cameras, the goal is to produce scene reconstructions that are both geometrically accurate and temporally stable. A particular focus is placed on characterizing the trade-off between these two desiderata - a gap that is obscured by conventional per-frame evaluation protocols - and on understanding which design choices drive or reduce temporal inconsistency.

The main contributions of this work are:

- **Evaluation framework.** A benchmarking pipeline for sparse multi-view video depth estimation that measures both per-frame spatial accuracy and sequence-level temporal consistency across multiple datasets

  Datasets require citation. Datasets subject to change

  (DexYCB, Hi4D, 4D-Dress, MultiCamRobolab).

- **Consistency metric.** A quantitative measure of the spatial– temporal error gap,

  $$\Delta_{\text{consistency}} = \text{Chamfer}_{4D} - \text{Chamfer}_{3D},$$

  which captures how much per-frame alignment masks underlying temporal instability (formally introduced in Chapter 4).

- **Empirical analysis.** A systematic comparison of spatial methods (MASt3R, VGGT) and temporal methods (VGGT4D, Mono4D) across two to four camera configurations, exposing failure modes and trade-offs not visible under standard benchmarks.
- **Alignment and refinement study.** An investigation of temporal alignment strategies and a global pose-graph refinement approach aimed at improving reconstruction stability without sacrificing spatial fidelity.

1.3 Thesis Outline

Chapter 3 reviews prior work on depth estimation, multi-view reconstruction, and dynamic scene modeling. Chapter 4 describes the datasets, metrics, and alignment strategies used in this study. Chapter 5 presents quantitative and qualitative results. Chapter 6 analyzes the key findings and limitations, and Chapter 8 summarizes the contributions and outlines directions for future work.

2

Background Knowledge

3
Related Work

4

Methodology

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Experiments

6
Limitations

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Future Work

8
Conclusions

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List of Acronyms

A

Artifacts Availability

The artifacts associated with this thesis can be found in the ifi-GitLab[1]. Within this repository, there are two main folders: one containing the code to fine-tune the model, and the other containing the code to generate the training data, including all the different datasets. The README file provided in the repository contains step-by-step instructions on how to set up the environment used in this thesis, install all necessary dependencies, and run the source code.

Upload your Code to GitLab rather than attaching it to the PDF.

[1]https://gitlab.ifi.uzh.ch/ddis/Students/Theses/<NAME>

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Declaration of independence for written work

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Zürich, April 17, 2026

Signature of
Fabio Di Meo